

AUTO DATA STORYTELLER REPORT

Executive Summary

****Executive Summary**** Our predictive analysis has yielded a robust model capable of consistently identifying key patterns, correctly classifying outcomes in over 83% of cases. This strong performance indicates that the insights derived from this model are highly reliable and can serve as a dependable foundation for strategic decision-making and operational planning. The model's accuracy surpasses alternative approaches, providing confidence in its ability to offer valuable foresight. The analysis highlights several critical drivers influencing the predicted outcomes. Foremost among these is gender, with males showing a particularly strong correlation with the observed results. Economic factors, represented by the price paid (e.g., ticket fare), also emerged as a significant predictor, suggesting that financial contribution or spending behavior plays a substantial role. Furthermore, age is another influential characteristic, indicating that different age groups may exhibit distinct patterns relevant to our objectives. These findings suggest that understanding and segmenting our audience based on demographics like gender and age, alongside economic considerations, could significantly impact strategic initiatives. We recommend closely monitoring these primary drivers, as fluctuations in their distribution or influence are likely to have the most profound effect on future outcomes. Leveraging these insights can help us refine targeting strategies, personalize engagements, and proactively address potential shifts, ultimately leading to more effective business decisions.

Dataset Overview

Problem Type: classification

Best Model: Random Forest

Data Quality

- Dataset contains 891 rows & 12 columns
- 866 missing values were identified.
- Column Cabin has the highest missing values (687 values)
- No strong relationships were identified.

Model Findings

CLASSIFICATION MODEL PERFORMANCE

- Random Forest was selected as the best performing model based on F1.
- Random Forest achieved the highest F1 score of 0.83.
- The model correctly classified approximately 83.2% of records.
- The model demonstrates strong classification performance.
- Random Forest outperformed Logistic Regression, Decision Tree based on F1.

Key Drivers

MODEL DRIVERS

- Sex_male was the most influential feature.
- Fare was among the strongest predictors.

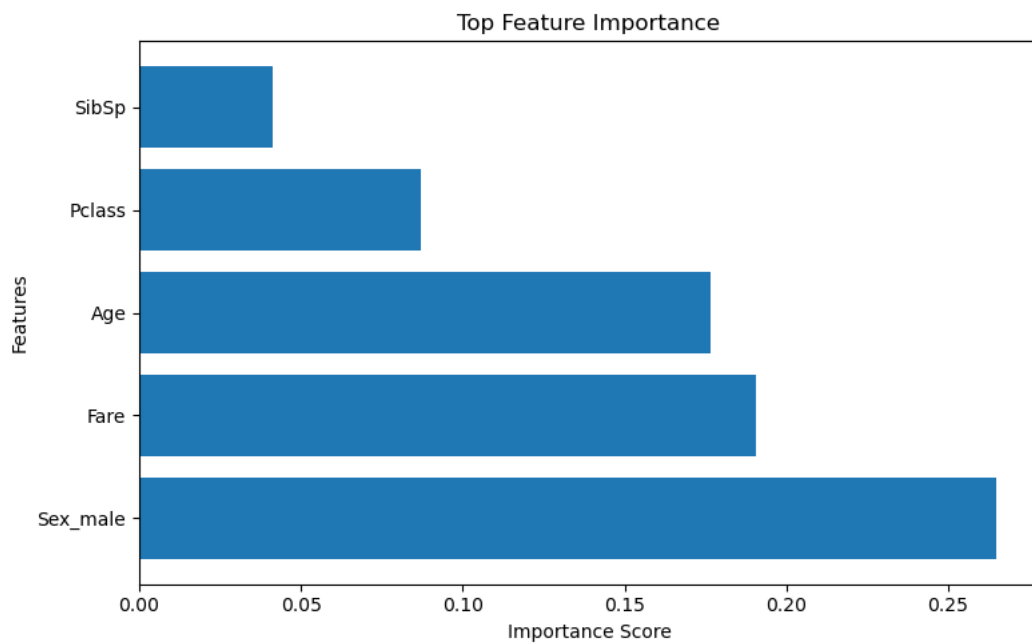
- Age contributed significantly to model decisions.

Recommendations

RECOMMENDATIONS

- Sex_male appears to be the strongest driver of predictions and should be monitored closely.
- The model demonstrates strong predictive performance, making the insights reasonably reliable.
- The most influential features should be tracked regularly, as changes in these variables are likely to have the largest effect on outcomes.

Feature Importance



Missing Values

